

Foundation (Jalapeno)

Q1. What is the inequality for 'x is greater than 2'?

- (A) $x < 2$
 (B) $x \leq 2$
 (C) $x > 2$
 (D) $x \geq 2$
 (E) $x = 2$

(A) $x \geq 1$ and $y \leq 2$

(B) $x \geq 1$ and $y < 2$

(C) $x > 1$ and $y \leq 2$

(D) $x \leq 1$ and $y \geq 2$

(E) $x < 1$ and $y > 2$

Q2. Solve $2x + 3y \geq 12$ for y:

- (A) $y \geq -\frac{2}{3}x + 4$
 (B) $y \leq -\frac{2}{3}x + 4$
 (C) $y \geq \frac{2}{3}x + 4$
 (D) $y \leq \frac{2}{3}x - 4$
 (E) $y = -\frac{2}{3}x + 4$

Q6. A factory produces two products (x and y). If $x + 2y \leq 120$, $x \geq 20$, $y \geq 20$, and x must be an integer, what is the maximum x value if y = 40?

(A) $x = 40$

(B) $x = 30$

(C) $x = 20$

(D) $x = 10$

(E) $x = 60$

Q3. Which point (x, y) satisfies $x + 2y \leq 4$ and $x - y \geq 1$?

- (A) $(-1, -1)$
 (B) $(1, 1)$
 (C) $(2, 1)$
 (D) $(1, 2)$
 (E) $(3, 1)$

Q7. Which inequality has the solution region between $y = 2x - 3$ and $y = -3x + 4$?

(A) $-3x + 4 \leq y \leq 2x - 3$

(B) $2x - 3 \leq y \leq -3x + 4$

(C) $2x - 3 \leq y \leq 3x + 4$

(D) $-3x + 4 \leq y \leq 3x + 4$

(E) $2x + 3 \leq y \leq -3x - 4$

Q4. If $3x + 4y \leq 24$, and x must be greater than 0, which is the maximum value of y for x = 4?

- (A) $y = 3$
 (B) $y = 2$
 (C) $y = 1$
 (D) $y = 0$
 (E) $y = -3$

Q8. What are the boundary lines of the solution set of $3x + 4y \geq 12$ and $y \leq 2x - 5$?

(A) $y = -\frac{3}{4}x + 3$, $y = 2x - 5$

(B) $y = -\frac{4}{3}x + 4$, $y = 2x + 5$

(C) $y = -\frac{3}{4}x + 3$, $y = 2x + 5$

(D) $y = \frac{3}{4}x - 3$, $y = -2x - 5$

(E) $y = -\frac{3}{4}x + 3$, $y = -2x + 5$

Q5. What system of inequalities has the solution set shown?

Open Ended Questions (FRQ)

Q9. A company has two factories. Factory 1 produces $3x$ units of a product. Factory 2 produces $2y$ units of the product. Production costs for Factory 1 and 2 are $x^2 + 2y^2$ and $y^2 + 3x^2$, respectively. The total production cost must not exceed \$240, the product must be greater than or equal to 20. Write down the inequalities representing these constraints.

Q10. If $x + 2y \leq 8$ and $x - 2y \geq -4$, and x and y must both be non-negative, find the values of x and y that maximize $x + y$. Write your final answer as an ordered pair (x, y) .

Chef's Tips

- Tip 1: Ensure all answers fit within given spaces.
- Tip 2: Provide clear step-by-step work for Free Response Questions.
- Tip 3: Use a ruler for graphing